

INNOVA JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
in preparation for General Certificate of Education Advanced Level  
**Higher 2**

CANDIDATE  
NAME

CLASS

INDEX NUMBER

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**BIOLOGY**

**9648/03**

Paper 3 Applications Paper

**25 August 2016**

**2 hours**

Additional Materials: Answer Paper  
Cover Page

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**READ THESE INSTRUCTIONS FIRST**

Write your name and class on all the work you hand in.  
Write in dark blue or black pen on both sides of the paper.  
You may use a soft pencil for any diagrams, graphs or rough working.  
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [ ] at the end of each question or part question.

| For Examiner's Use |    |
|--------------------|----|
| 1                  | 16 |
| 2                  | 13 |
| 3                  | 11 |
| 4                  | 12 |
| 5                  | 20 |
| Total              | 72 |

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This document consists of **16** printed pages.



**Section A**Answer **all** questions.

- 1** Diabetes mellitus is a disease that results in the inability to regulate one's blood glucose levels, either due to lack of insulin or insensitivity to insulin.

Injection of insulin before meals helps patients to keep blood glucose within set point. Insulin used to be purified from animal sources but genetic engineering has now provided an alternative source.

- (a)** State **two** advantages of genetically engineered insulin compared to those obtained from animal sources.

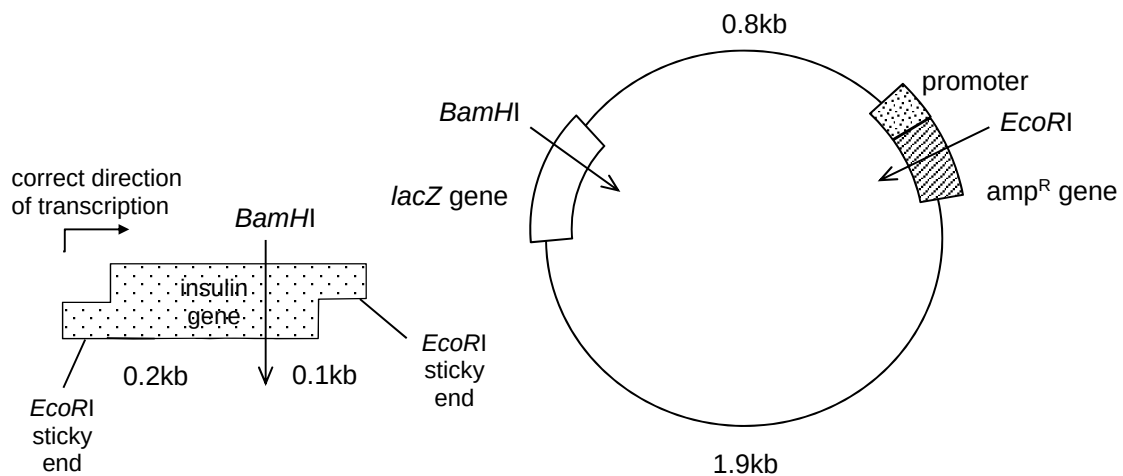
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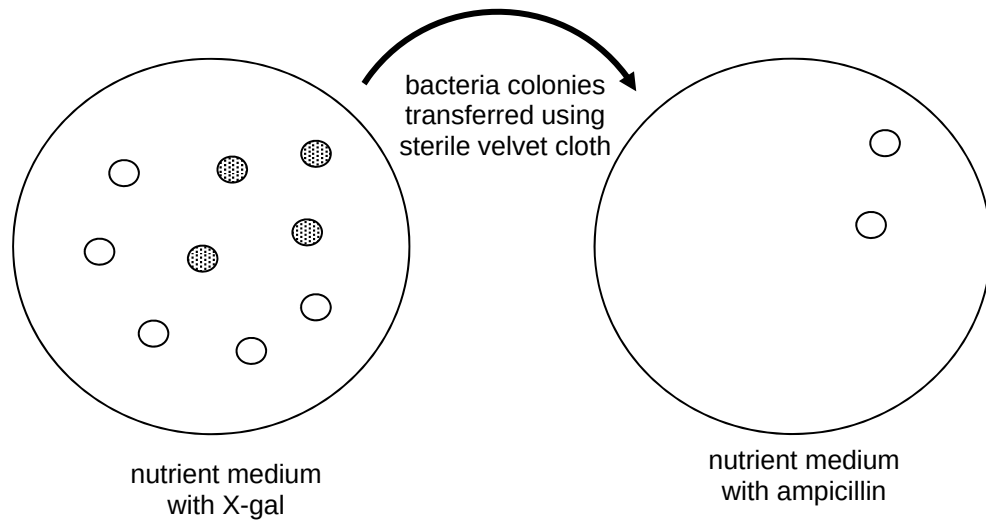
..... [2]

Human insulin is isolated using *EcoRI* from the cDNA library and ligated into plasmid shown in Fig. 1.1. The number of base pairs between the restriction recognition sites is shown.



**Fig. 1.1**

Plasmids were introduced into bacteria host cells and grown on nutrient medium containing X-gal. A replica plate was made by transferring bacteria colonies from the first plate to nutrient medium containing ampicillin. Colonies growing in the two plates are shown in Fig. 1.2.



**Fig. 1.2**

Legend

- blue colony
- white colony

**(b) (i)** State **one** possible bacteria host cell.

[1]

**(ii)** Explain how plasmids were introduced into host cells.

[4]

(c) (i) On Fig. 1.2, label the recombinant colony/colonies with an 'X'. [1]

(ii) Explain how you arrive at your answer in (c)(i).

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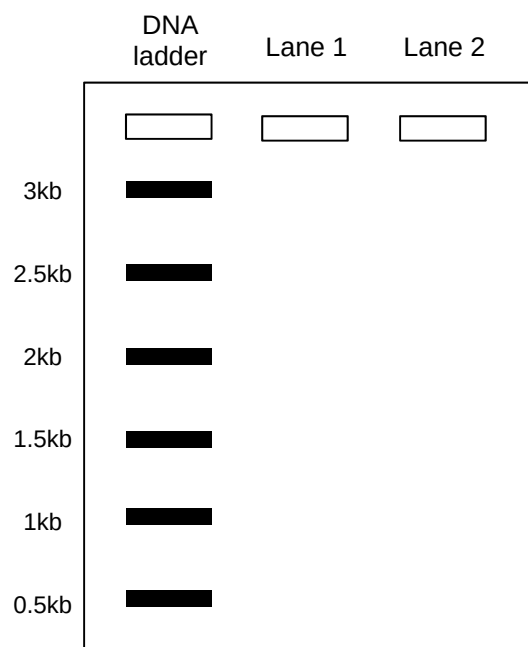
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----- [2]

The insulin gene was found to be inserted into plasmids in two possible orientations. To determine if the insulin gene was inserted correctly, recombinant plasmids were isolated, digested with *Bam*HI and separated by gel electrophoresis.

Fig. 1.3 shows the outline of the electrophoresed gel.



**Fig. 1.3**

(d) On Fig. 1.3, illustrate the bands found if plasmids with insulin gene inserted

(i) correctly was loaded in lane 1; [1]

(ii) wrongly was loaded in lane 2. [1]

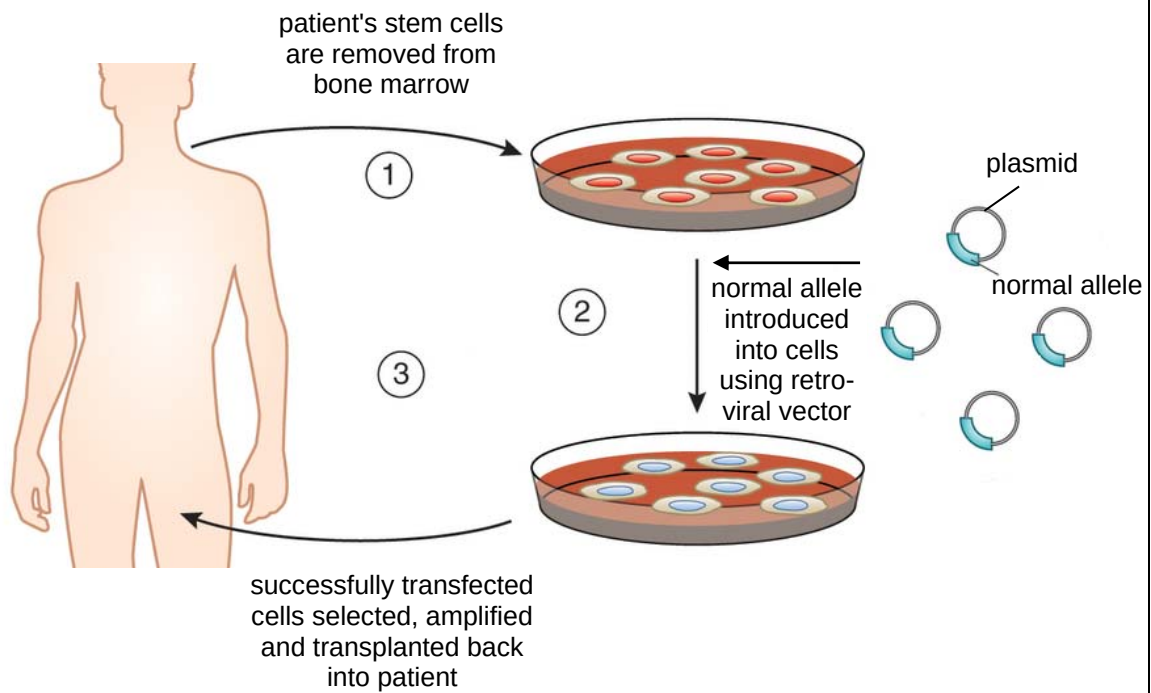
[4]

2 X-linked Severe Combined Immunodeficiency (X-SCID) is a recessive genetic disorder. Diseased individuals suffer from compromised immune system due to malfunctioning lymphocytes.

[3]

[1]

X-SCID can be treated by *ex vivo* gene therapy shown in Fig. 2.1.



**Fig. 2.1**

- (b)** (i) State **two** reasons why the *ex vivo* approach was used instead of *in vivo*.

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..... [2]

- (ii) Suggest why the normal allele was ligated into a plasmid before introduction into cells.

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..... [1]

- (iii) Suggest how successfully transfected cells were selected.

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..... [2]

Normal lymphocytes were detected in some patients' blood for several months following gene therapy. Some of the patients who underwent this treatment were subsequently diagnosed to suffer from leukaemia (cancer of blood cells).

- (c) (i) Explain why normal lymphocytes were detected in patients' blood for several months following gene therapy.

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- (ii) Explain how gene therapy could have led to the development of leukaemia in other patients.

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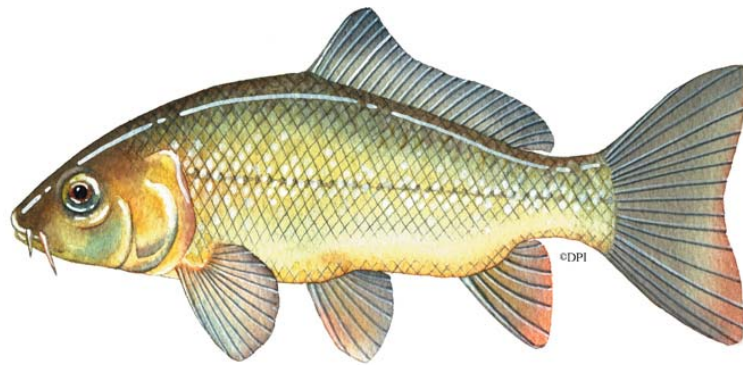
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[Total: 13]

- 3 The carp, shown in Fig. 3.1, is a large freshwater fish native to central Asia. It was introduced to Australia by humans in the mid-1800s, where it subsequently became an aquatic pest.



**Fig. 3.1**

- (a) Suggest why the carp became a pest when it was introduced into Australia.

[1]

Scientists tried to control the carp by using gene technology. They inserted a gene into carp embryos that prevents the production of an enzyme, aromatase. Aromatase catalyses the conversion of testosterone into oestrogen. The inserted gene causes female embryos to develop as males.

The inserted gene produces mRNA with a base sequence that is complementary to the base sequence of the mRNA produced by the aromatase gene. Table 3 shows part of the base sequence of the aromatase gene. It also shows part of the base sequence of the mRNA produced by the inserted gene.

**Table 3**

|                                      | base sequence |   |   |   |   |   |
|--------------------------------------|---------------|---|---|---|---|---|
| template strand of aromatase gene    | T             | G |   |   | A |   |
| mRNA transcribed from aromatase gene |               |   |   |   |   |   |
| template strand of inserted gene     |               |   |   |   |   |   |
| mRNA transcribed from inserted gene  |               |   | G | C |   | U |

- (b) Complete Table 3 to show the base sequences in the mRNAs transcribed and the inserted gene. [3]



(c) Describe how the inserted gene was introduced into the carp embryos.

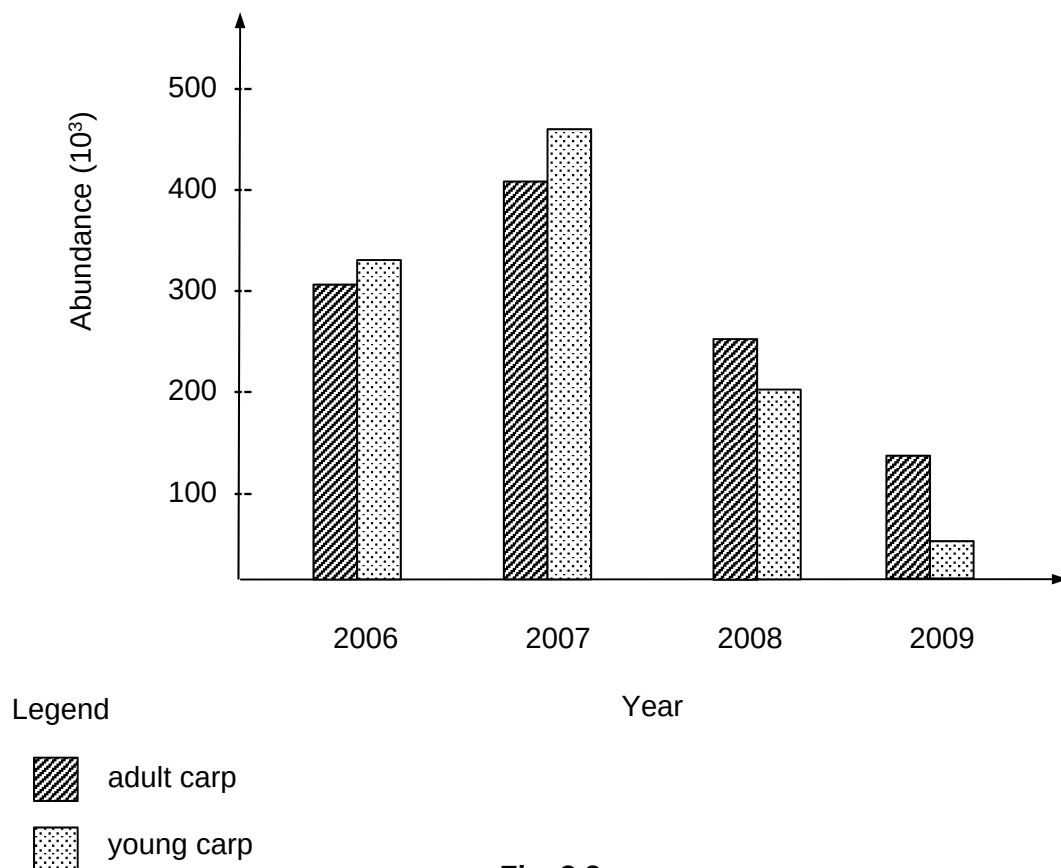
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..... [2]

The genetically modified (GM) carp were released into the environment in 2006. Fig. 3.2 shows the effect on wild carp populations for the subsequent years.



**Fig. 3.2**

- (d) With reference to Fig. 3.2 and information on the inserted gene, explain the effect of releasing GM carp on carp populations.

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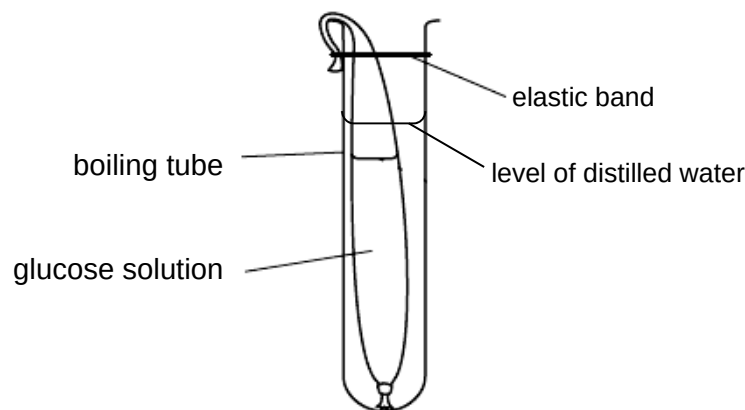
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..... [5]

[Total: 11]

#### 4 Planning question

Visking tubing is a selectively permeable membrane that allows small molecules like glucose to pass through. Fig. 4.1 shows how a Visking tubing can be set up to investigate diffusion rates.



**Fig. 4.1**

Using this information and your own knowledge, design an experiment to investigate the effect of concentration on rate of glucose diffusion.

You must use:

- 10% glucose solution,
- distilled water,
- Benedict's solution,
- Visking tubing,
- boiling tube,
- elastic band.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- syringes,
- Bunsen burner, tripod stand.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagrams, if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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This image shows a full page of a worksheet designed for handwriting practice. It consists of approximately 20 horizontal dashed lines spaced evenly across the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

[illegible]

## 5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

- (a)** The patient's own stem cells are frequently modified for use in treating genetic diseases via gene therapy.

Explain why these stem cells are suited for this purpose. [6]

- (b)** The therapeutic gene used for gene therapy can be amplified using polymerase chain reaction (PCR). Describe the process of PCR. [8]

- (c)** The therapeutic gene can be delivered using viral or non-viral vectors. Compare the use of these two modes of gene delivery. [6]

[Total: 20]